

Air University



Digital Logic Design (Lab)

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Experiment No. 10

Title: Implementation of a Seven-Segment Display

Objectives

- Understand how segments are arranged in a seven-segment display.
- Manually control the display segments using direct connections and resistors.
- Confirm the display's functionality by lighting specific segments for the digits 0–9.

Equipment and Components

- Proteus Software (for simulation)
- Logic Trainer Kit (for hands-on experimentation)
- Common anode seven-segment display
- 6V external battery
- Standard current-limiting resistors
- Connecting wires (for direct segment control)
- Breadboard

Introduction

A seven-segment display is a device used to visually represent numeric values (0–9) through seven individual LED segments. In this experiment, we will operate the display by linking its segments to a power source using resistors, avoiding the need for a decoder IC, such as the 7447. Each segment, labeled from a to g, can be manually activated through wiring, allowing us to control specific segments of the display. This hands-on approach not only highlights the physical layout of the display but also underscores the critical role of resistors in limiting current to protect the LEDs. The experiment provides practical wiring techniques and illustrates how illuminating segments correspond to numeral representation.

Procedure

A. Proteus Simulation

Circuit Design:

- Add a common cathode seven-segment display in the Proteus software.
- Connect each segment (a–g) to a single cell battery.
- Simulate the digits by manually lighting up the segments; for example, to display "0," activate segments a, b, c, d, e, and f.

Observation:

- Confirm that the correct segments light up for each digit (0–9) as per the truth table.

B. Logic Trainer Implementation

Hardware Setup:

- Secure the common anode display to the breadboard.
- Position resistors between each segment pin (a–g) and the positive terminal of the external battery.
- Ground the common anode pin of the display.
- Use wires to connect or disconnect the segments from the negative terminal of the battery to turn them on.

Testing:

- For each digit (0–9), connect the corresponding segments to the battery through the resistors.
- Observe the lit segments and compare them against the truth table.

Truth Table

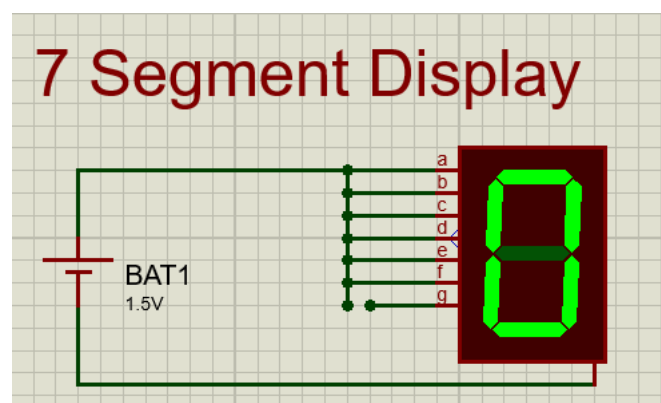
The following truth table outlines which segments illuminate on the seven-segment display:

Display	Segments						
	a	b	c	d	e	f	g
0	1	1	1	1	1	1	0
1	0	1	1	0	0	0	0
2	1	1	0	1	1	0	1
3	1	1	1	1	0	0	1
4	0	1	1	0	0	1	1
5	1	0	1	1	0	1	1
6	1	0	1	1	1	1	1
7	1	1	1	0	0	0	0
8	1	1	1	1	1	1	1
9	1	1	1	1	0	1	1

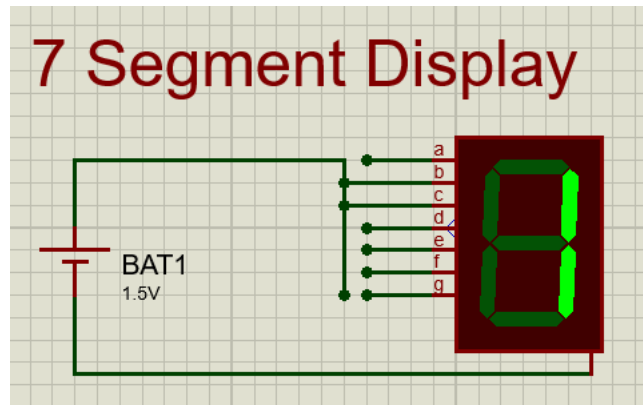
In this table, '1' indicates that the segment is **ON**, while '0' signifies that it is **OFF**.

Simulated Results

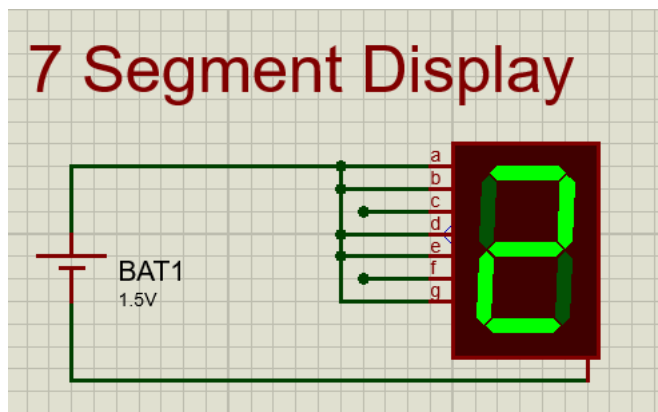
a=1, b=1, c=1, d=1, e=1, f=1, g=0 → Displays '0'



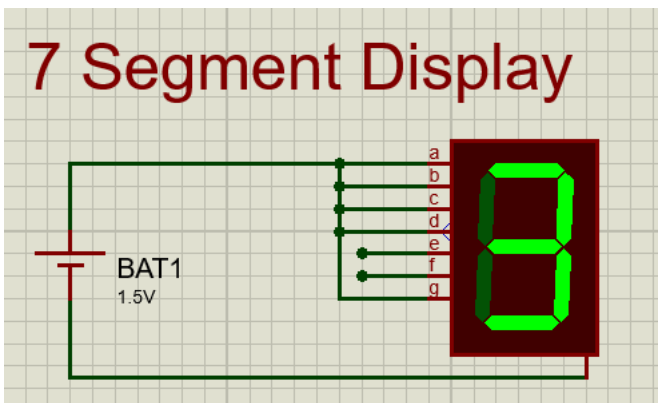
$a=0, b=1, c=1, d=0, e=0, f=0, g=0 \rightarrow$ Displays '1'



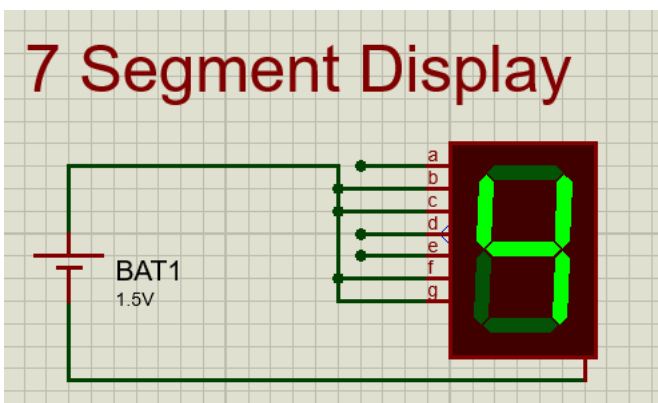
$a=1, b=1, c=0, d=1, e=1, f=0, g=1 \rightarrow$ Displays '2'



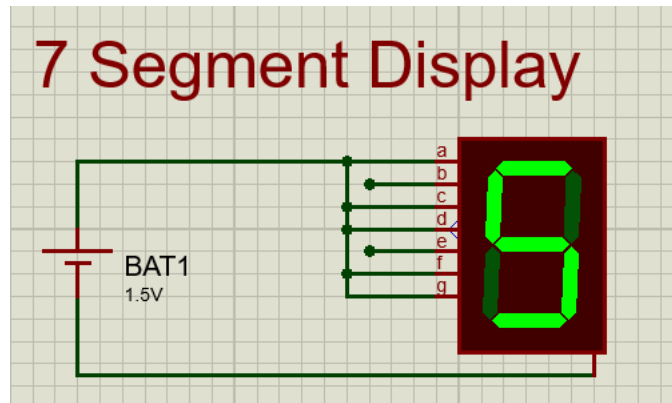
$a=1, b=1, c=1, d=1, e=0, f=0, g=1 \rightarrow$ Displays '3'



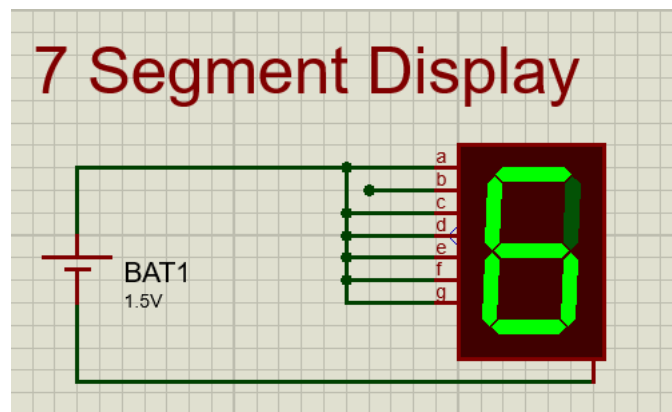
$a=0, b=1, c=1, d=0, e=0, f=1, g=1 \rightarrow$ Displays '4'



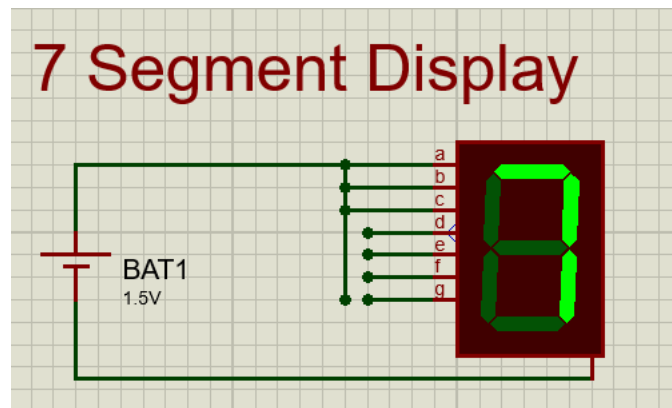
a=1, b=0, c=1, d=1, e=0, f=1, g=1 → Displays '5'



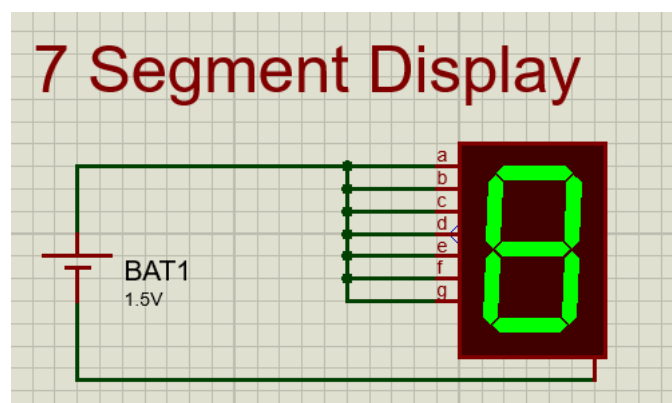
a=1, b=0, c=1, d=1, e=1, f=1, g=1 → Displays '6'



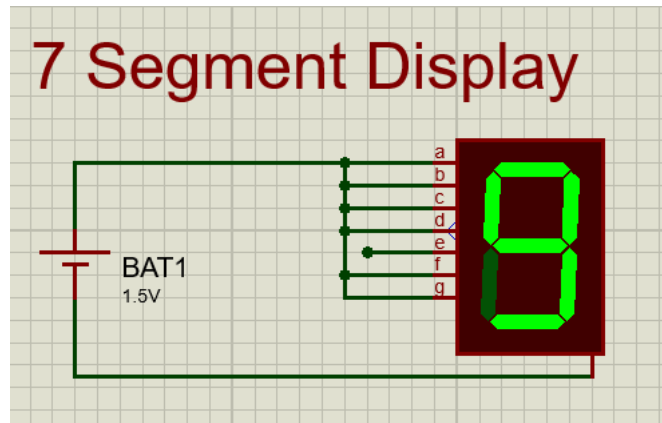
a=1, b=1, c=1, d=0, e=0, f=0, g=0 → Displays '7'



a=1, b=1, c=1, d=1, e=1, f=1, g=1 → Displays '8'



a=1, b=1, c=1, d=1, e=0, f=1, g=1 → Displays '9'



Conclusions

This experiment demonstrated how to directly manipulate a seven-segment display using manual connections and resistors. By selectively energizing specific segments through direct links, we were able to clearly observe the connection between which segments lit up and the corresponding numeral displayed. Using standard resistors helped keep current levels safe, protecting the LED segments from possible damage while ensuring they remained bright and visible. This hands-on approach underscored the importance of understanding the display's structure, as well as the role of current-limiting components in practical circuits. While this method effectively supports educational purposes, it requires precise wiring for each numeral, making it less efficient for automated systems.

THE END
